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SUBJECT CODE NO: RNP-8015
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (Civil Engg.) (NEP) Sem - III
Examination May/June-2026
Strength of Material

[Time: 1:30 Hours]

[Max. Marks: 30]

Please check whether you have got the right question paper.

N. B

1. Part A: Question No 1 is compulsory.
2. Attempt any TWO questions from remaining Part B: Q.2 to Q.5.

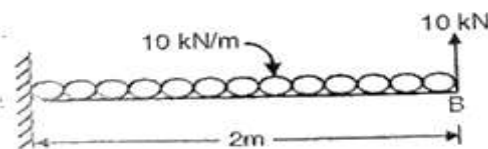
PART – A**Q1 Solve any 5 following questions****10**

- a) Define strain and write its mathematical expression.
- b) What is modulus of rigidity?
- c) Define thermal stress.
- d) What is point of contra flexure?
- e) Define polar moment of inertia.
- f) What is effective length of column?
- g) State Euler's theory of columns.

PART – B**Q.2 Solve any TWO from the following****10**

- 1) A compound bar consists of a steel rod and a brass rod rigidly connected together. The system is subjected to an axial load of 120 kN. Determine the stresses developed in each material.
 Take: Area of steel rod = 900 mm², Area of brass rod = 1200 mm²,
 $E_s = 200 \text{ GPa}$, $E_b = 100 \text{ GPa}$

- 2) Draw SFD and BMD of given beam.

10

- 3) T Section with flange (320 mm × 80 mm) and web (100 mm × 80 mm), is used as a cantilever beam of 3.6 m span subjected to a u.d.l. of intensity 180 kN/m over its full span. Determine the maximum stresses in the beam.

10

- 4) Analyse the given truss by joint method. Also draw the nature of forces in each member. **10**

